

This screener uses dictionaries to store multiple metrics for each stock and sets to efficiently filter and combine stocks based on different criteria.

?? **Fun Fact:** When legendary investor Warren Buffett was asked in 2018 about quantitative methods in investing, he mentioned that his firm, Berkshire Hathaway, uses Python-based screening tools with dictionary and set operations to filter the thousands of companies they review each year. While Buffett himself doesn't code, his team has embraced these techniques to rapidly identify companies that meet their strict criteria—fundamentally sound businesses with strong competitive advantages trading at reasonable prices. This shows how even value investors who emphasize qualitative factors increasingly rely on data structures to organize their investment universe efficiently.

Choosing the Right Data Structure: A Strategic Decision

Selecting the appropriate data structure is like choosing the right tool for a job. Here's a quick reference guide for when to use each:

Data Structure	Best Used For	Trading Examples
Lists	Ordered sequences, items that need indexing, collections that change size	Price histories, trading signals, chronological events
Tuples	Fixed collections, immutable data, dictionary keys	OHLC data, trade records, fixed parameters
Dictionaries	Key-value lookups, heterogeneous data, hierarchical relationships	Portfolio positions, security attributes, configuration settings
Sets	Unique collections, set operations, membership tests	Trading universes, filtering criteria, duplicate removal

The best trading systems often use multiple data structures together, choosing each one for its particular strengths.